SYSTEM AND METHOD FOR ADAPTIVE QUANTUM-SAFE SECURITY AND PRIVACY FOR HEALTHCARE DATA

(CADA-AASM) — DRAFT PATENT SPECIFICATION

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October 2026

Draft for discussion with a registered patent agent. It is not legal advice. A limited, abstract-level literature search (11 October 2026; see `docs/LITERATURE_REVIEW.md`) was performed and is reflected in Section 2.1, but no formal patent prior-art search has been performed*; every statement of novelty is the inventor's position and must be verified by a registered patent agent before filing.*

1. Field of the Invention

The invention relates to information security for networked medical devices and health-record systems, and in particular to a controller that jointly selects the strength of a hybrid post-quantum transport-cryptography suite and the strength of data-privacy mechanisms (differential privacy and (K, L)-anonymisation) as a function of a measured real-time latency budget, subject to a minimum protection tier for raw clinical data.

2. Background

Post-quantum primitives standardised by NIST (ML-KEM, ML-DSA) have public keys, ciphertexts and signatures one to two orders of magnitude larger than X25519/Ed25519 equivalents; for example an ML-DSA-87 signature is 4 627 bytes against 64 bytes for Ed25519, and spans several Ethernet frames. Streaming Internet-of-Medical-Things (IoMT) telemetry such as ICU electrocardiography is subject to an end-to-end latency service-level agreement (SLA) of about 200 ms. A statically configured high-security suite can therefore violate the SLA under congestion. Separately, data released for secondary use must be de-identified, using differential privacy for numeric streams and (K, L)-anonymisation for tabular records, and these mechanisms also consume time and affect utility. In the inventor's review, transport cryptography and data-privacy mechanisms are configured independently of one another and of live latency conditions (but see Section 2.1).

2.1 Known approaches (limited search, abstract level)

Runtime selection of post-quantum parameters from network conditions is *known*: a TLS 1.3 client that selects among hybrid ML-KEM groups from a measured round-trip time and device limits (AdaptiveQKE, 2025–26, not peer-reviewed); a context-aware optimiser that selects lattice-, code- or hash-based PQC configurations for 6G vehicular links from predicted channel state (CAAP, arXiv:2602.01342); and an IoMT aggregation framework using Q-learning for key management (Q-PRADAX, Sci. Rep. 15:37176, 2025). Closed-loop adaptation of a differential-privacy budget from channel and resource conditions is also *known* (ALPINE, arXiv:2510.17162, mobile edge crowdsensing). In the works read, none couples a post-quantum transport suite and data-privacy parameters (differential privacy and (K, L)-anonymisation) through a single latency budget, and none applies a minimum protection floor to raw clinical data while adapting a de-identified stream. **Consequently the inventive concept should not be framed as "adaptive PQC selection" per se; it resides in the joint coupling of transport and privacy parameters under one budget, together with the dual-domain floor.** Dependent claims on the persistence filter, make-before-break re-keying and batch post-quantum signing are individually weak and are best treated as supporting embodiments.

3. Summary

A gateway measures a round-trip time and a processor utilisation, computes a latency budget as the service-level latency minus the round-trip time minus a fraction of the utilisation, and maps the budget through a steering matrix to one of a plurality of tiers. Each tier *simultaneously* fixes (a) a key-encapsulation parameter set, (b) a signature parameter set, (c) a symmetric AEAD cipher, (d) a differential-privacy parameter ε, and (e) anonymisation parameters K and L. A persistence filter commits a tier change only after the same different target tier is produced on a predetermined number of consecutive measurement cycles. A domain-policy layer applies the selected tier unchanged to a de-identified-data stream but clamps the tier applied to a raw-protected-health-information stream to be no lower than a predetermined floor tier.

4. Brief Description of the Drawings

FIG. 1 control loop; FIG. 2 steering matrix and hysteresis state machine; FIG. 3 hybrid handshake; FIG. 4 packet format with per-packet classical signature and per-epoch post-quantum signature; FIG. 5 dual-domain policy; FIG. 6 operator console. (Drawings to be prepared by the patent draftsman from `docs/ARCHITECTURE.md`.)

5. Detailed Description

5.1 Measurement and budget

A prober measures round-trip time R (ms) over a socket to a peer and processor utilisation C (percent). The latency budget is $L = S - R - \alpha C$ with $S = 200\,\text{ms}$ and $\alpha = 0.5$ in the preferred embodiment. In an alternative embodiment R and C are smoothed by an exponentially weighted moving average before use.

5.2 Steering matrix

In the preferred embodiment: $L > 80$ ms selects Tier 5 (ML-KEM-1024, ML-DSA-87, AES-256-GCM, $\varepsilon = 0.1$, $K = 5$, $L_{div} = 3$); $40 < L \leq 80$ selects Tier 3 (ML-KEM-768, ML-DSA-65, AES-256-GCM, $\varepsilon = 0.5$, $K = 3$, $L_{div} = 2$); $L \leq 40$ selects Tier 1 (ML-KEM-512, ML-DSA-44, ChaCha20-Poly1305, $\varepsilon = 1.0$, $K = 2$, $L_{div} = 2$). The thresholds and parameter sets are exemplary.

5.3 Persistence filter

The controller stores a pending tier and a counter. When the target equals the active tier the counter is cleared. When the target differs it is compared with the pending tier: if equal the counter is incremented, otherwise the pending tier is replaced and the counter set to one. When the counter reaches N (three in the preferred embodiment) the active tier is replaced and the counter cleared.

5.4 Hybrid key establishment

Client and server perform an X25519 exchange and an ML-KEM encapsulation of the selected parameter set. The server signs the handshake transcript with both an Ed25519 and an ML-DSA key of the selected parameter set; the client accepts only if both verify. The master secret is derived by HKDF-SHA256 over the concatenation of the two shared secrets, salted with a hash of the transcript (an XOR combiner is an alternative embodiment).

5.5 Make-before-break re-keying

Upon a tier change the handshake of the new tier is executed on a control path while data continues on the existing session; the data path switches on completion.

5.6 Packet authentication

Each packet carries an AEAD-protected payload whose associated data include a tier identifier, session identifier and sequence number, and a classical signature. A post-quantum signature is computed over a chained hash of all packets of an epoch of E packets and attached to the epoch-closing packet, amortising the post-quantum signature size; $E = 1$ is a degenerate embodiment.

5.7 Dual-domain policy

The tier T_a selected by the controller is applied to the de-identified stream, where it also sets ε of a Laplace mechanism with sensitivity-scaled noise and the parameters (K, L_{div}) of a Mondrian multidimensional partitioning anonymiser. The tier applied to the raw stream is $T_c = \max(T_a, T_{floor})$ with $T_{floor} = 3$ in the preferred embodiment. A researcher-export function enforces $K \geq 5$ and $L_{div} \geq 3$ irrespective of T_a and verifies the released table independently before release.

5.8 Auditing

Control events are appended to a log in which each entry contains the hash of the previous entry and a keyed message-authentication code.

6. Claims (draft)

1. A computer-implemented method comprising: (a) measuring a round-trip time and a processor utilisation of a gateway handling medical telemetry; (b) computing a latency budget from a service-level latency, the round-trip time and a fraction of the processor utilisation; (c) selecting, from a plurality of tiers according to the latency budget, a tier that defines jointly a post-quantum key-encapsulation parameter set, a post-quantum signature parameter set, a symmetric cipher, a differential-privacy parameter and at least one anonymisation parameter; and (d) applying the selected tier to transport protection of a data stream and to de-identification of data derived from the stream.

2. The method of claim 1, wherein a change of the selected tier is committed only after the same new tier is selected on a predetermined number of consecutive measurement cycles.

3. The method of claim 1, wherein tiers differ in at least a ciphertext size of the key-encapsulation parameter set and a signature size of the signature parameter set, and a lower latency budget selects a tier with smaller sizes.

4. The method of claim 1, further comprising applying to a stream of raw protected health information a tier not lower than a predetermined floor tier irrespective of the latency budget, while applying the selected tier without such a floor to a de-identified stream.

5. The method of claim 1, wherein the key establishment combines a classical and a post-quantum shared secret by a key-derivation function and the handshake transcript is signed by both a classical and a post-quantum signature scheme, both of which must verify.

6. The method of claim 1, wherein upon a change of tier a handshake for the new tier is executed while data continues on the previous session.

7. The method of claim 1, wherein each packet carries a classical signature and a post-quantum signature is computed over a chained digest of a plurality of packets and carried on a single packet of the plurality.

8. The method of claim 1, wherein the anonymisation parameter comprises K and L_{div} of a multidimensional partitioning anonymiser and the differential-privacy parameter is the privacy-loss parameter of a Laplace mechanism applied to a continuous physiological signal, the parameters increasing in strength with the latency budget.

9. The method of claim 4, wherein an export to a research user enforces a minimum K and L_{div} independent of the selected tier and is refused if an independent verification of the released table fails.

10. The method of claim 1, wherein control events are recorded in a hash-chained log with a keyed message-authentication code per entry.

11. A system comprising a processor and memory storing instructions that, when executed, perform the method of any of claims 1–10.

12. A non-transitory computer-readable medium storing instructions that, when executed, perform the method of any of claims 1–10.

Drafting note for the patent agent: consider adding to claim 1 the dual-domain floor of claim 3 as a fall-back independent claim, since the joint coupling alone may be argued to be an obvious combination of the known adaptive-PQC and adaptive-privacy approaches of Section 2.1.

7. Abstract

A gateway for medical telemetry measures round-trip time and processor load, derives a latency budget, and selects a tier that jointly sets a hybrid post-quantum cipher suite, a differential-privacy parameter and anonymisation parameters, damped by a persistence filter and subject to a minimum tier for raw clinical data.